# Hassnain Ali

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## Profile

AI-focused Full Stack Developer with strong expertise in backend engineering, intelligent workflow automation, and production-grade web application development. Experienced in building scalable MERN applications integrated with Large Language Models (LLMs), AI APIs, cloud services, authentication systems, Redis caching, and asynchronous processing. Passionate about designing secure, high-performance software that combines modern backend architecture with practical AI solutions to solve real-world business problems..

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## Skills

**Backend:** Node.js, Express, REST APIs, JWT, OAuth, Middleware Architecture
**Frontend:** React, Vite, EJS, Tailwind, JavaScript, HTML, CSS
**Database:** MongoDB, Mongoose, Cloud Storage (Superbase / S3-compatible systems)
**DevOps & Cloud:** Docker, Docker Compose, Vercel, Git, GitHub
**Security:** Helmet.js, Express Validator, HPP, CORS, Rate Limiting, Secure Cookies, Bcrypt, CSRF Protection
**AI-Assisted Development:** Google Gemini, Claude AI, GitHub Copilot, OpenRouter, Prompt Engineering, AI Workflow Automation, AI API Integration, AI Content Generation.

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## Education

**BS Software Engineering**
Virtual University of Pakistan

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## Experience

**Full Stack Developer — SolnetInfo**
**Intern —** *Feb 2025 – Jul 2025*
**Junior Developer —** *Aug 2025 – Mar 2026*

- Developed and maintained RESTful APIs and backend services using Node.js and Express.js
- Designed MongoDB schemas, implemented CRUD operations, and optimized database queries for performance and scalability
- Built authentication and authorization systems using JWT, secure session handling, and role-based access control
- Integrated Redis caching to reduce database load and improve application response times
- Structured backend applications using modular architecture, reusable services, and middleware-based routing patterns
- Collaborated on frontend development using React.js and Tailwind CSS to deliver responsive user interfaces
- Performed debugging, testing, validation, and performance optimization across full-stack applications
- Contributed to production-level codebases by implementing new features, resolving bugs, and improving maintainability
- Dockerized applications to ensure consistent development, testing, and deployment environments
- Applied security best practices including input validation, error handling, and API protection mechanisms

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## Projects

### AI Email Campaign Platform

**Live:** https://ai-email-compaign-agent-project-fro.vercel.app

- Developed an AI-powered email campaign platform that generates, reviews, and optimizes marketing emails using Large Language Models (LLMs).
- Integrated Google Gemini and OpenRouter to create personalized email content, improve readability, detect spam risks, and enhance deliverability.
- Engineered asynchronous email processing using Redis and BullMQ for scalable background job execution.
- Built campaign management, contact management, scheduling, analytics dashboard, and delivery tracking.
- Integrated multiple email providers including SendGrid, Brevo, Mailgun, and Postmark with secure API configuration.
- Implemented campaign analytics, contact management, CSV import/export, OAuth authentication, secure REST APIs, and production-ready deployment.

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### AI WhatsApp & E-Commerce Assistant Platform

**Live:** https://fyp-xi-flame.vercel.app

- Developed an AI-powered communication platform integrating web chat, WhatsApp Business, and email into a unified customer support solution.
- Built an intelligent assistant capable of answering customer queries, retrieving product information, and providing store assistance using Large Language Models.
- Integrated e-commerce functionality enabling AI-powered product search, pricing, inventory lookup, and customer guidance across web and WhatsApp.
- Implemented secure authentication, persistent conversation history, session management, and modular REST APIs.
- Designed scalable backend architecture supporting AI workflows, third-party integrations, and multi-channel communication.
- Integrated cloud-based AI services with backend APIs to automate customer interactions, reduce manual support effort, and improve response efficiency.

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### GalleryPro

**Live:** https://gallery-project-frontend.vercel.app

- Developed a production-ready MERN image management platform with secure authentication, cloud storage, and enterprise-grade backend architecture.
- Integrated Redis caching and Supabase Storage to optimize data retrieval, image management, and application performance.
- Implemented Google OAuth, JWT authentication, refresh token workflow, and role-based authorization for secure user access.
- Built advanced features including drag-and-drop uploads, image optimization, favorites, filtering, search, and responsive UI.
- Dockerized the application and designed scalable REST APIs following modular architecture and modern backend engineering practices.
- Implemented modular backend architecture with reusable middleware, centralized validation, and secure API design for maintainability and scalability.

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### Google Drive Enterprise

**Live**: https://google-drive-ecru.vercel.app

- Engineered a full-stack Google Drive clone supporting secure file storage, folder hierarchy, multi-file uploads, and enterprise-grade file management.
- Built scalable backend services integrating MongoDB Atlas, Redis caching, and Supabase Storage to improve performance and optimize cloud-based file handling.
- Implemented secure authentication using JWT, refresh token rotation, CSRF protection, input validation, and role-based authorization.
- Designed advanced file management capabilities including recursive folder navigation, bulk downloads, favorites, indexed search, and automated trash lifecycle management.
- Developed a production-ready backend architecture focused on scalability, security, caching strategies, and efficient cloud-native deployment.
- Built efficient file indexing and metadata management to enable fast search, filtering, and navigation across large file collections.
- Designed cloud-native storage workflows supporting secure uploads, bulk downloads, automatic file lifecycle management, and scalable object storage integration.

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### Engineering Highlights

- Designed scalable RESTful API architectures following modular backend design principles.
- Built AI-powered applications integrating Google Gemini, OpenRouter, and multiple third-party AI services.
- Developed asynchronous processing systems using Redis caching and BullMQ background workers.
- Implemented secure authentication with JWT, refresh tokens, OAuth 2.0, and role-based authorization.
- Engineered cloud-native applications using MongoDB Atlas, Supabase Storage, and Vercel deployment.
- Integrated multi-channel communication systems combining AI, WhatsApp Business, email services, and e-commerce workflows.
- Built production-ready Docker environments for consistent local development and deployment.
- Applied modern backend security practices including Helmet, HPP, Rate Limiting, CORS, CSRF Protection, Express Validator, and Bcrypt.
- Experienced with AI-assisted software development using Claude, GitHub Copilot, Google Gemini, and modern prompt engineering techniques.
- Focused on building maintainable, scalable, and high-performance software following production-level engineering standards.

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### Additional Software Projects

- **Full-Stack Applications:** E-commerce Platform, Authentication System, Task Management System, Notes Application, and additional CRUD-based web applications.

- **AI & API Integrations:** AI Chatbot, Weather Dashboard, Currency Converter, and several applications utilizing REST APIs and third-party services.

- **Frontend Development:** Responsive UI implementations inspired by Amazon, Daraz, Walmart, eBay, and AliExpress using modern HTML, CSS, and JavaScript.

- **JavaScript Applications:** Interactive browser applications and games including Tic-Tac-Toe and Rock-Paper-Scissors, demonstrating proficiency in JavaScript, event-driven programming, and responsive UI development.